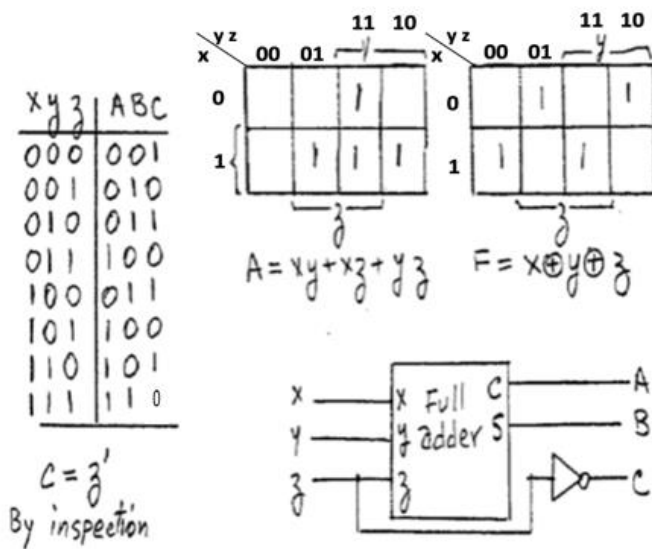


HW5 Solutions Spring 2025

1. Problem 1-16

Design a combination circuit with three inputs x, y, z and three outputs A, B, C . When the binary input is 0, 1, 2, or 3, the binary output is one greater than the input. When the binary input is 4, 5, 6, or 7, the binary output is one less than the input.

Solution:



2. Problem 1-19

A sequential circuit has two D flip-flops A and B, two inputs x and y, and one output z. The flip-flop input equations and the circuit output are as follows:

$$D_A = x'y + xA$$

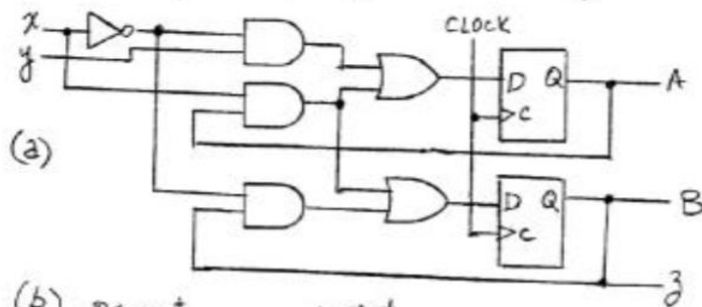
$$D_B = x'B + xA$$

$$Z = B$$

- Draw the logic diagram of the circuit
- Tabulate the state table

Solution:

1-19 $D_A = x'y + xA$; $D_B = x'B + xA$; $z = B$



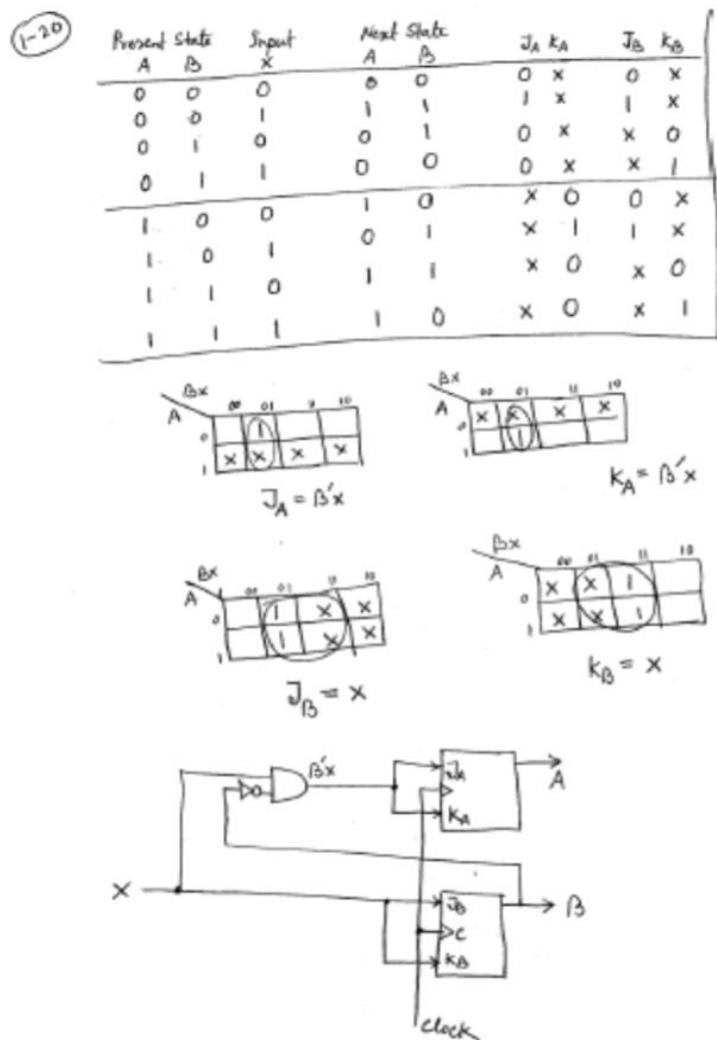
(b)

Present state AB	Inputs x y	Next state AB	Output z
00	00	00	0
00	01	10	0
00	10	00	0
00	11	00	0
01	00	01	1
01	01	11	1
01	10	00	1
01	11	00	1
10	00	00	0
10	01	10	0
10	10	11	0
10	11	11	0
11	00	01	1
11	01	11	1
11	10	11	1
11	11	11	1

3. Problem 1-20

Design a 2-bit count-down counter. This is a sequential circuit with two flip-flops and one input x . When $x = 0$, the state of the flip-flops does not change. When $x = 1$, the state sequence is 11, 10, 01, 00, 11, and repeat.

Solution:



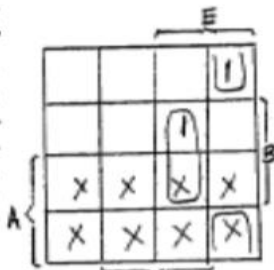
4. Problem 1-21

Design a sequential circuit with two JK flip-flops A and B and two inputs E and x. if $E = 0$, the circuit remains in the same state regardless of the value of x. When $E = 1$ and $x = 1$, the circuit goes through the state transitions from 00 to 01 to 10 to 11 back to 00, and repeat. When $E = 1$ and $x=0$, the circuit goes through the state transitions from 00 to 11 to 10 to 01 back to 00 and repeat.

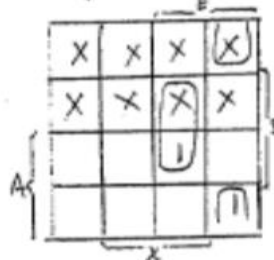
Solution:

1-21 Count up-down binary counter with enable E

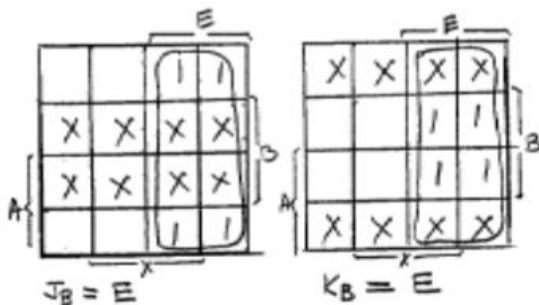
Present state		Inputs		Next state		Flip-flop inputs			
A	B	E	x	A	B	J_A	K_A	J_B	K_B
0	0	0	0	0	0	0	x	0	x
0	0	0	1	0	0	0	x	0	x
0	0	1	0	1	1	1	x	1	x
0	0	1	1	0	1	0	x	1	x
0	1	0	0	0	1	0	x	x	0
0	1	0	1	0	1	0	x	x	0
0	1	1	0	0	0	0	x	x	1
0	1	1	1	1	0	1	x	x	1
1	0	0	0	1	0	x	0	0	x
1	0	0	1	1	0	x	0	0	x
1	0	1	0	0	1	x	1	1	x
1	0	1	1	1	1	x	0	1	x
1	1	0	0	1	1	x	0	x	0
1	1	0	1	1	1	x	0	x	0
1	1	1	0	1	0	x	0	x	1
1	1	1	1	0	0	x	1	x	1



$$J_A = (Bx + B'x')E$$



$$K_A = (Bx + B'x')E$$



$$J_B = E$$

$$K_B = E$$

5. A burglar alarm for a bank is designed so that it senses four input signal lines. Line *A* is from the secret control switch, line *B* is from a pressure sensor under a steel safe in a locked closet, line *C* is from a battery-powered clock, and line *D* is connected to a switch on the locked closet door. The following conditions produce a logic 1 voltage on each line.

A: The control switch is closed.

B: The safe is in its normal position in the closet.

C: The clock is between 10:00 and 14:00 hours (banking hours).

D: The closet door is closed.

Write the equation of the control logic for the burglar alarm that produces a logic 1 (rings a bell) when the safe is moved and the control switch is closed, or when the closet is opened after banking hours, or when the closet is opened with the control switch open.

Solution:

A: The control switch is closed.

B: The safe is in its normal position in the closet.

C: The clock is between 10:00 and 14:00 hours.

D: The closet door is closed.

Logic 1 (rings a bell) when:

i) The safe is moved and the control switch is closed = AB'

ii) Or when the closet is opened after banking hours = $C'D'$

iii) Or when the closet is opened with the control switch open = $A'D'$

$$F(A,B,C,D) = AB' + C'D' + A'D'$$

6. Mr. and Ms. Boren have two children, John and Jane. When eating out they will go to a restaurant that serves only hamburgers or one that serves only chicken. Before going out, the family votes to decide on the restaurant. The majority wins, except when Mom and Dad agree, and in that case, they win. Any other tie votes produce a trip to the chicken restaurant. Design a logic circuit that will automatically select the restaurant when everyone votes.

Solution:

Let's assume,

1 = hamburgers

0 = chicken

i) The majority wins, except when Mom and Dad agree

- ii) If Mom and Dad agree, they win.
- iii) Any other tie votes produce a trip to the chicken = 0

A	B	C	D	F
Dad	Mom	John	Jane	Vote
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

The Logic Function: $F(A, B, C, D)$

$$= A'BCD + AB'CD + ABC'D' + ABC'D + ABCD' + ABCD$$

$$= A'BCD + AB'CD + ABC'(D+D') + ABC(D'+D)$$

$$= A'BCD + AB'CD + ABC' + ABC$$

$$= A'BCD + AB'CD + AB(C' + C)$$

$$= A'BCD + AB'CD + AB$$

$$= AB + AB'CD + A'BCD$$

$$= A(B + B'CD) + A'BCD$$

$$= A(B + CD) + A'BCD$$

$$= AB + ACD + A'BCD$$

$$= AB + CD(A + A'B)$$

$$= AB + CD(A + B)$$

$$= AB + ACD + BCD$$

7. In most automobiles, the lights, radio, and power windows operate only if the ignition switch is turned on. In this case, the ignition switch acts as an “enabling” signal.

Design the circuit to model this automotive subsystem using the following variables and definitions:

Ignition switch *IG*: Value 0 if off and value 1 if on
 Light switch *LS*: Value 0 if off and value 1 if on
 Radio switch *RS*: Value 0 if off and value 1 if on
 Power window switch *WS*: Value 0 if off and value 1 if on
 Lights *L*: Value 0 if off and value 1 if on
 Radio *R*: Value 0 if off and value 1 if on
 Power windows *W*: Value 0 if off and value 1 if on

Solution:

Table contains the condensed truth table for the operation of this automobile subsystem. Note that when the ignition switch *IS* is off (0), all of the controlled accessories are off (0) regardless of their switch settings. This is indicated by the first row of the table. With the use of X's, this condensed truth table with just nine

TABLE
Truth Table for Enabling Application

Input Switches				Accessory Control		
IS	LS	RS	WS	L	R	W
0	X	X	X	0	0	0
1	0	0	0	0	0	0
1	0	0	1	0	0	1
1	0	1	0	0	1	0
1	0	1	1	0	1	1
1	1	0	0	1	0	0
1	1	0	1	1	0	1
1	1	1	0	1	1	0
1	1	1	1	1	1	1

rows represents the same information as the usual 16-row truth table. Whereas X's in output columns represent don't-care conditions, X's in input columns are used to represent product terms that are not minterms. For example, 0XXX represents the product term $\overline{IS}$. Just as with minterms, each variable is complemented if the corresponding bit in the input combination from the table is 0 and is not complemented if the bit is 1. If the corresponding bit in the input combination is an X, then the variable does not appear in the product term. When the ignition switch *IS* is on (1), then the accessories are controlled by their respective switches. When *IS* is off (0), all accessories are off. So *IS* replaces the normal values of the outputs *L*, *R*, and *W* with a fixed value 0 and meets the definition of an *ENABLE* signal.

8. Design a circuit that counts the number of 1's present in 3 inputs A , B and C . Its output is a two-bit number X_1X_0 , representing that count in binary.

a) Write the truth table for this circuit.

b) Find the minimized logic equations for outputs X_1 and X_0 using Boolean algebra and K-map. Determine the one gives better simplification for X_1 and X_0 .

c) Draw the corresponding logic diagram for this circuit. Label all inputs and outputs.

Solution:

a)

A	B	C	X_1	X_0
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

b)

A K-map for X_0 is not very helpful, as it results in 4 isolated minterms, $\overline{A}\overline{B}C + \overline{A}B\overline{C} + A\overline{B}\overline{C} + ABC$. However, we may recall from the binary adder lab that this equation can be factored further using the XOR operator $\oplus$.

Factor out $\overline{A}$ and A from their respective terms to obtain:

$$X_0 = \overline{A}(\overline{B}C + B\overline{C}) + A(\overline{B}\overline{C} + BC) = \overline{A}(B \oplus C) + A(\overline{B} \oplus \overline{C}) = A \oplus (B \oplus C).$$

$$X_0 = A \oplus B \oplus C$$

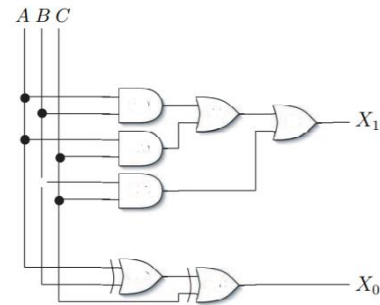
K-Map for X_1 :

$A \backslash C$	00	01	11	10
0	0	0	1	0
1	0	1	1	1

SOP expression for X_1 : blue vertical pair= BC
green horizontal pair= AC , red horizontal pair= AB

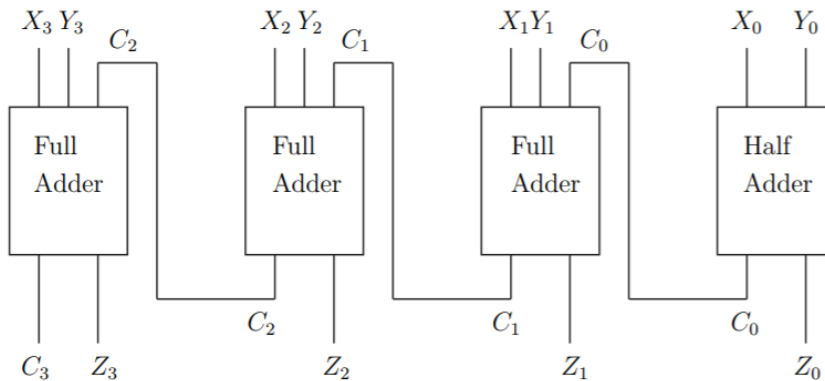
$$X_1 = AB + AC + BC$$

c)



9. Draw a block diagram of a 4-bit adder, using one 1-bit half and three 1-bit full adders. Do not draw a gate-level diagram. Show and label all inputs and outputs.

Solution:



10. There are three inputs A , B , and C (least significant bit) and one output Y of a digital circuit. The inputs represent a 3-bit unsigned number. The output Y is a 1 if and only if the input number is a multiple of 2 or 3 but not both.

- Draw the truth table
- Draw the K-map for Y
- Determine the simplified Boolean function Y .

Solution:

a)

Truth Table				
	A	B	C	Y
0	0	0	0	0
1	0	0	1	0
2	0	1	0	1
3	0	1	1	1
4	1	0	0	1
5	1	0	1	0
6	1	1	0	0
7	1	1	1	0

b)

Karnaugh Map

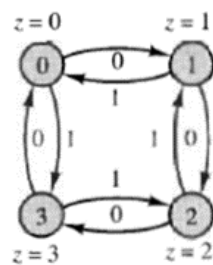
		AB			
C		00	01	11	10
	0	0	1	0	1
	1	0	1	0	0

c) Boolean Expression: $Y = A'B + AB'C'$

11. Design an up/down wraparound counter with four states (0, 1, 2, 3) using clocked JK flip-flops. A control signal x is to be used as follows: When $x = 0$ the circuit counts forward (up); when $x = 1$, backward (down) (Note: You need to draw the state diagram, truth table for JK flipflop inputs, and also find expressions of the flipflops by using Karnaugh map):-

Solution:

State diagram of the counter is as follows.



State Table:

Present State		Input
A	B	x
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

Next State	
A	B
0	1
1	1
1	0
0	0
1	1
0	1
0	0
1	0

JK flip-flop inputs			
J_A	K_A	J_B	K_B
0	x	1	x
1	x	1	x
1	x	x	1
0	x	x	1
x	0	1	x
x	1	1	x
x	1	x	1
x	0	x	1

J_A

	B			
	0	1	0	1
A {	x	x	x	x

X

$$J_A = B'x + Bx'$$

K_A

	B			
	x	x	x	x
A {	0	1	0	1

X

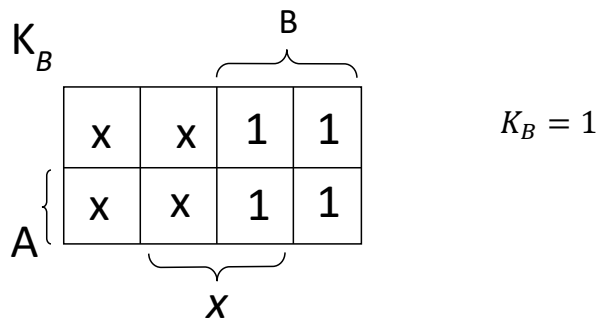
$$K_A = B'x + Bx'$$

J_B

	B			
	1	1	x	x
A {	1	1	x	x

X

$$J_B = 1$$



12. Design a binary counter that goes through the sequence 0 1 2 3 4 5 6 7 8 9 0 using four JK flip-flops. [Find the expressions for all the JK flipflops using the Karnaugh map.](#) Note that there will be a number of don't care conditions in your design.

Solution:

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Present State				Next State			
D	C	B	A	D	C	B	A
0	0	0	0	0	0	0	1
0	0	0	1	0	0	1	0
0	0	1	0	0	0	1	1
0	0	1	1	0	1	0	0
0	1	0	0	0	1	0	1
0	1	0	1	0	1	1	0
0	1	1	0	0	1	1	1
0	1	1	1	1	0	0	0
1	0	0	0	1	0	0	1
1	0	0	1	0	0	0	0
1	0	1	0	X	X	X	X
1	0	1	1	X	X	X	X
1	1	0	0	X	X	X	X
1	1	0	1	X	X	X	X
1	1	1	0	X	X	X	X
1	1	1	1	X	X	X	X

JK flip-flop inputs							
J _D	K _D	J _C	K _C	J _B	K _B	J _A	K _A
0	x	0	x	0	x	1	x
0	x	0	x	1	x	x	1
0	x	0	x	x	0	1	x
0	x	1	x	x	1	x	1
0	x	x	0	0	x	1	x
0	x	x	0	1	x	x	1
0	x	x	0	x	0	1	x
1	x	x	1	x	1	x	1
x	0	0	x	0	x	1	x
x	1	0	x	0	x	x	1
x	x	x	x	x	x	x	x
x	x	x	x	x	x	x	x
x	x	x	x	x	x	x	x
x	x	x	x	x	x	x	x
x	x	x	x	x	x	x	x
x	x	x	x	x	x	x	x

DC \ BA	BA			
	00	01	11	10
00	0	0	0	0
01	0	0	1	0
11	X	X	X	X
10	X	X	X	X

$$J_D = CBA$$

DC \ BA	BA			
	00	01	11	10
00	X	X	X	X
01	X	X	X	X
11	X	X	X	X
10	0	1	X	X

$$K_D = A$$

DC \ BA	BA			
	00	01	11	10
00	0	0	1	0
01	X	X	X	X
11	X	X	X	X
10	0	0	X	X

DC \ BA	BA			
	00	01	11	10
00	X	X	X	X
01	0	0	1	0
11	X	X	X	X
10	X	X	X	X

$$J_c = K_c = BA$$

From the above K-map we can get:

$$J_D = CBA \quad K_D = A$$

$$J_C = K_C = BA$$

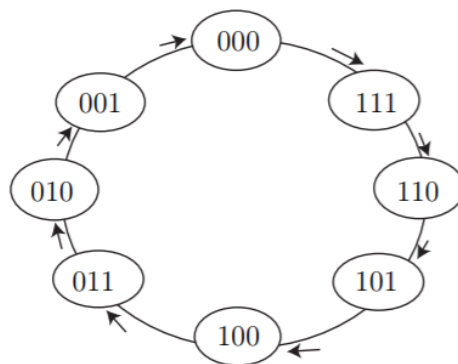
Similarly:

$$J_B = D'A \quad K_B = A$$

$$J_A = KA = 1$$

13. We want to design a 3-bit binary wraparound down-counter using J-K flip-flops. Draw the state diagram, considering the 3-bit flip-flop outputs as the state.

Solution:



14. A sequential circuit has two D Flip-Flops (A and B), two inputs (x and y), and one output (z). The circuit is specified by the following next-state and output equations:

$$A(t+1) = x'y + xA$$

$$B(t+1) = x'B + xA$$

$$z = B$$

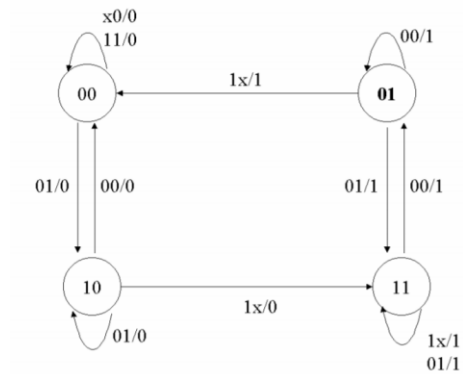
- a) List the state table for the sequential circuit.
b) Draw the corresponding state diagram.

Solution:

a)

Present State		Inputs		Next State		Output
A	B	x	y	A	B	z
0	0	0	0	0	0	0
0	0	0	1	1	0	0
0	0	1	0	0	0	0
0	0	1	1	0	0	0
0	1	0	0	0	1	1
0	1	0	1	1	1	1
0	1	1	0	0	0	1
0	1	1	1	0	0	1
1	0	0	0	0	0	0
1	0	0	1	1	0	0
1	0	1	0	1	1	0
1	0	1	1	1	1	0
1	1	0	0	0	1	1
1	1	0	1	1	1	1
1	1	1	0	1	1	1
1	1	1	1	1	1	1

b)



x represents 0 or 1.

15. Design a sequential circuit with two D Flip-Flops, A and B, and one input x. When x = 0, the state of the circuit remains the same. When x = 1, the circuit goes through the state transitions from 00 to 01 to 11 to 10 back to 00, and repeats. [Find the expressions for the D flipflops using the Karnaugh map.](#)

Solution:

Present State AB	Input x	Nest State AB
00	0	00
00	1	01
01	0	01
01	1	11
10	0	10
10	1	00
11	0	11
11	1	10

$$A \updownarrow$$

A $\updownarrow$